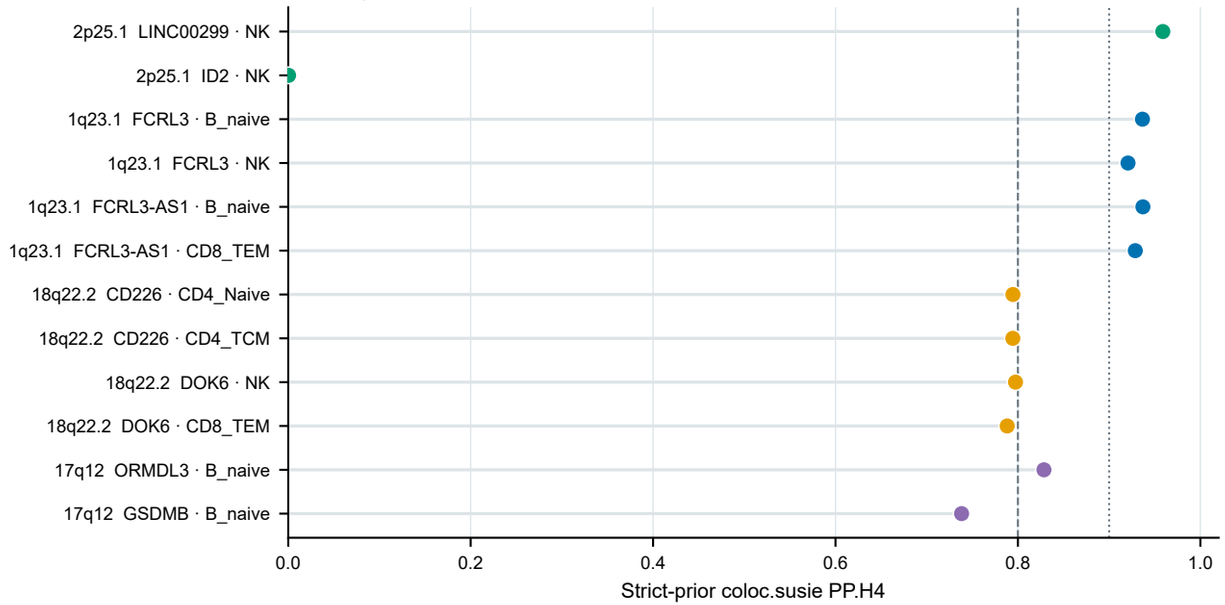
Supplementary Figure S3 | Competing genes and locus-level downgrading

Strict-prior signal sharing is separated from transcript resolution and independent disease replication.

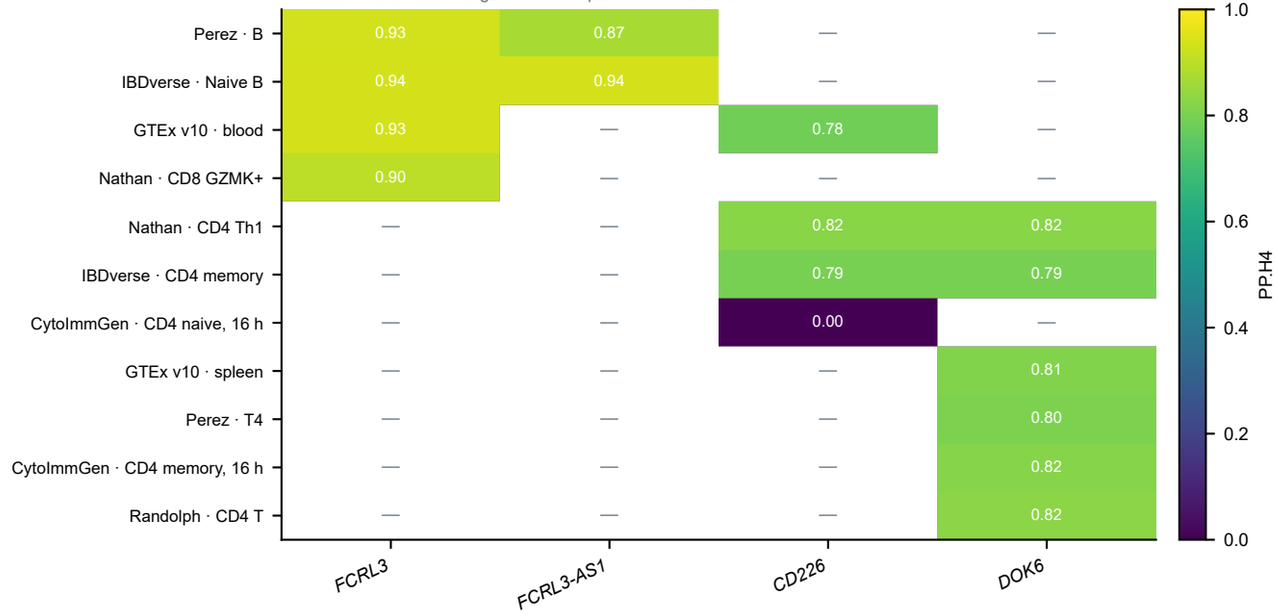
A Within-locus competitors

Multiple cell combinations at one locus are correlated evidence



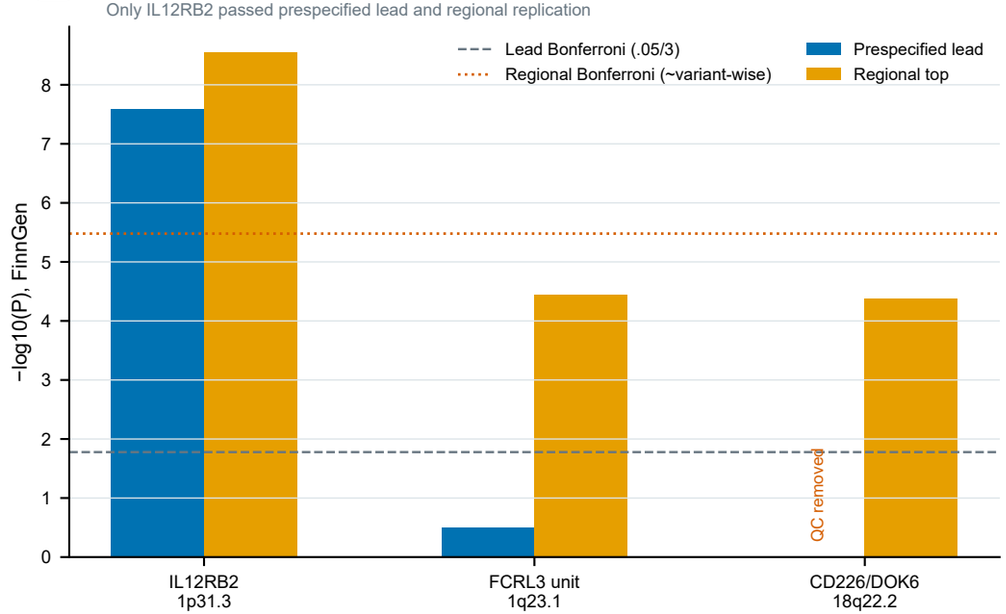
B Study-level external eQTL audit

Blank cells mean the gene–context pair was not available



C Independent disease replication

Only IL12RB2 passed prespecified lead and regional replication



D Evidence lock and downgrade rationale

Gene nomination is bounded by resolution, replication and druggability

2p25.1 — **LINC00299 > ID2**
Strong contrast | PP.H4 0.959 vs 0.0004
Boundary: No validated therapeutic target

1q23.1 — **FCRL3 / FCRL3-AS1**
Regulatory unit | External molecular support; FinnGen P=0.321
Boundary: Sense/antisense transcript unresolved

18q22.2 — **CD226 / DOK6**
Candidate unit | PP.H4 0.788–0.797; proxies P=0.296–0.467
Boundary: Lead QC-removed; causal gene unresolved

17q12 — **ORMDL3 / GSDMB**
Ambiguous | PP.H4 0.829 vs 0.738
Boundary: Same signal cannot rank causal gene

Interpretation boundary: repeated gene–cell combinations at the same locus are not independent discoveries; PP.H4 supports signal sharing, not therapeutic causality.